

CHAPTER – VI

INTERFACE:

- A interface in java is a blueprint of a class.
- It contains abstract method(without body) that must be implemented by any class that chooses to use it.

WHY DO WE NEED INTERFACE:

- Achieve Multiple Inheritance
- To Achieve Abstraction

SYNTAX:

```
interface InterfaceName{  
    //constance variable  
    //abstract method  
}
```

EXAMPLE:

```
interface Father{  
    void fat();  
}  
interface Mother{  
    void mot();  
}  
class Child implements Father, Mother{  
    public void fat() {  
        System.out.println("Ram is my father ");  
    }  
}
```

```
    }  
    public void mot() {  
        System.out.println("Seetha is my mother");  
    }  
}  
public class MultipleInhe{  
    public static void main(String [] args) {  
        Child c = new Child();  
        c.fat();  
        c.mot();  
    }  
}
```

FINAL KEYWORD:

- Final keyword is a non -access modifier used to restrict changes to variables, methods and class.
- Once a final keyword is declared it cannot be modified, overridden or inherited.

TYPES:

- Final variable – Aadhar Card
- Final method – Company's core policy
- Final class – Sealed box

EXAMPLE:

```
class ConstantInterest{
    final int int_rate = 10;
    double prin= 10000;
    double time = 3;
    double interest;
    ConstantInterest(){
        interest=(prin*int_rate*time)/100;
    }
    void display() {
        System.out.println("The Constant Rate:"+interest);
    }
}

public class ConstantIntFinal {
    public static void main(String [] args) {
        ConstantInterest c1 = new ConstantInterest();
        c1.display();
    }
}
```

INSTANCEOF KEYWORD:

- instanceof keyword is a comparison operator used to test whether an object is an instance of a specific class or subclass.
- Checks relationship between object and class.
- It returns boolean value.

SYNTAX:

objectName instanceof ClassName

EXAMPLE:

```
class Animal1{
    void display() {
        System.out.println("The Animal Class");
    }
}

class Dog1 extends Animal1{
    void show() {
        System.out.println("The Dog Class");
    }
}

public class InstOfEg {
    public static void main(String [] args) {
        Dog1 d1 = new Dog1();
        d1.display();
        d1.show();
        System.out.println("Checking that Dog is instance of
Animal");
        System.out.println(d1 instanceof Animal1);
    }
}
```

THIS KEYWORD:

- This keyword refers to the current object of the class.

WHY DO WE USE THIS KEYWORD:

- A method needs access to the entire object, not just one variable.
- We want to send the current object to another class or method.
- Object to object interaction.

EXAMPLE:

```
class Stu {  
    String name;  
    Stu(String name){  
        this.name=name;  
        System.out.println("The name of the Student :"+name);  
    }  
}  
  
class Teach1 {  
    String sub;  
    Teach1(String sub, String name){  
        this.sub=sub;  
        System.out.println("Name of the teacher:"+name+"  
Subject:"+sub);  
    }  
}
```

```
public class ThisConst{  
    public static void main(String [] args) {  
        Stu s1= new Stu("Anu");  
        Teach1 t1 = new Teach1("Abi","Maths");  
    }  
}
```